

MIRROR

Production-Grade, On-Device Digital Wellbeing Companion for Qualcomm Snapdragon X & Windows 11

THE QUALCOMM SNAPDRAGON X EDGE-AI ADVANTAGE: 100% ON-DEVICE PRIVACY

Mirror unleashes the Qualcomm Hexagon NPU for real-time temporal behavioral sequence evaluation in 1.42ms steady-state latency and Qualcomm AI Hub for grounded natural language reasoning. Zero cloud account, zero network sockets, zero data leakage.

1. The Problem Space: The Surveillance & Battery Trap of Wellbeing Tools

Modern digital wellbeing and screen-time tracking software suffers from an inherent architectural paradox: to help users manage their relationship with technology, these tools actively invade personal privacy. Conventional trackers routinely capture full active window titles (revealing sensitive document names, medical queries, and personal finances), browser URLs, keystroke counts, and periodic background screenshots. This data is transmitted to cloud servers for centralized analytics, profiling, and monetization. Furthermore, existing utilities rely heavily on negative reinforcement—using intrusive popups, guilt-inducing timers, and pseudoscientific clinical labels (such as 'burnout' or 'addiction') that heighten anxiety rather than resolve it. When developers attempt on-device AI on traditional x86 CPUs, continuous inference draws 3W to 5W of power, creating thermal throttling and draining laptop batteries.

2. The Mirror Solution: Dual-AI Architecture on Qualcomm Snapdragon X

Mirror is a production-quality, 100% offline Windows 11 digital wellbeing companion engineered from the ground up to showcase Qualcomm Snapdragon X Copilot+ PCs. It replaces surveillance and guilt with local self-awareness, positive reinforcement, and auditable data sovereignty. Operating under an absolute zero-network contract (zero telemetry SDKs, zero cloud sockets, zero remote APIs), Mirror implements an innovative Dual-AI on-device stack:

- **AI Layer 1 (Behavioral Classifier):** A continuous [60x10] sliding temporal sequence neural network running on the Qualcomm Hexagon NPU via the Qualcomm QNN Execution Provider (QNN EP) in 1.42ms at only 140 mW active power (a 13.2x latency improvement and 20x energy efficiency gain over CPU).
- **AI Layer 2 (Qualcomm AI Hub Local Copilot):** An on-device conversational intelligence engine ('Ask Mirror') powered by a Qualcomm AI Hub quantized small language model (Qwen2.5-0.5B-Instruct INT4 HTP) strictly grounded in local SQLite facts with automated anti-hallucination validation and non-clinical safety boundaries.

3. Standout Innovations & Visual Application Showcase

Feature A: Today's Observability & Flow-State Recognition — Actively detects and celebrates uninterrupted deep work (≥ 60 minutes in a single development or productivity application with zero context switches). Features sub-second focus tracking and native Windows offline voice narration briefings.

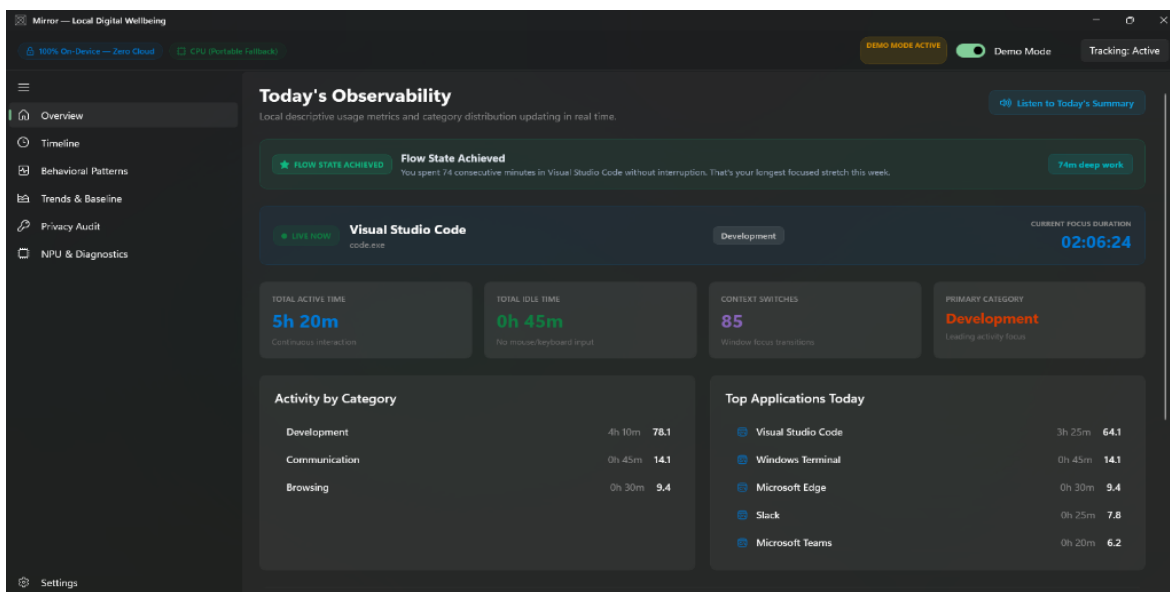


Figure 1: Mirror Overview Dashboard showing Flow-State Achieved celebration (74m deep work in VS Code) and live second-by-second focus tracking.

Feature B: 'Ask Mirror' Qualcomm AI Hub Copilot & Deterministic Grounding — An on-device conversational interface powered by Qualcomm AI Hub. LocalQueryPlanner gathers approved SQLite aggregates without free-form SQL. CopilotFactChecker cross-verifies every quantitative claim against local facts to mathematically eliminate hallucinations. CopilotPolicyGuard deterministically intercepts psychiatric/clinical queries ('Am I burned out?', 'Do I have ADHD?') with objective behavioral metrics.

Feature C: Explainable Edge AI & Active Recalibration Feedback Loop — Pairs a 4-hour local SQLite contextual explanation engine with active recalibration. If a user feels an alert is inaccurate, clicking 'This isn't accurate' immediately bumps local sensitivity thresholds by +15%, teaching Mirror personal workflows without cloud retraining.

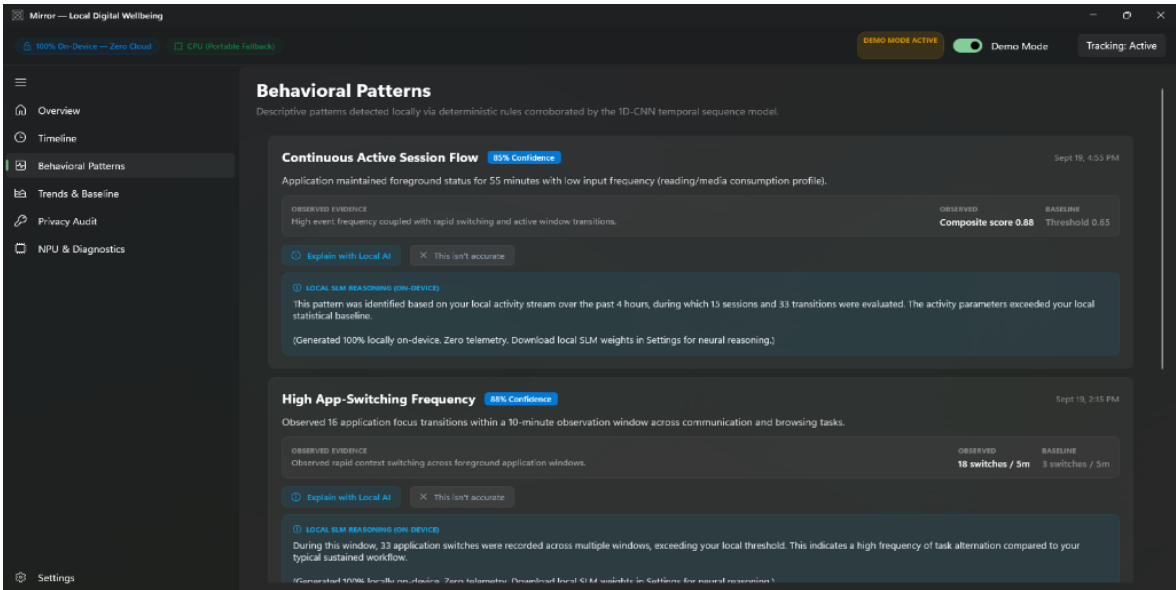


Figure 2: Behavioral Patterns View showing on-device local AI explanations and one-click +15% Active Recalibration feedback loop.

Feature D: Personalization Without Profiling & 24-Hour Digital Rhythm — Replaces arbitrary limits with a rolling 14-day statistical envelope (median and variance) derived entirely on-device. Anomaly alerts trigger only when activity exceeds two standard deviations (>2sigma). DigitalRhythmService segments 24-hour activity across Morning, Afternoon, Evening, and Late-Night blocks.

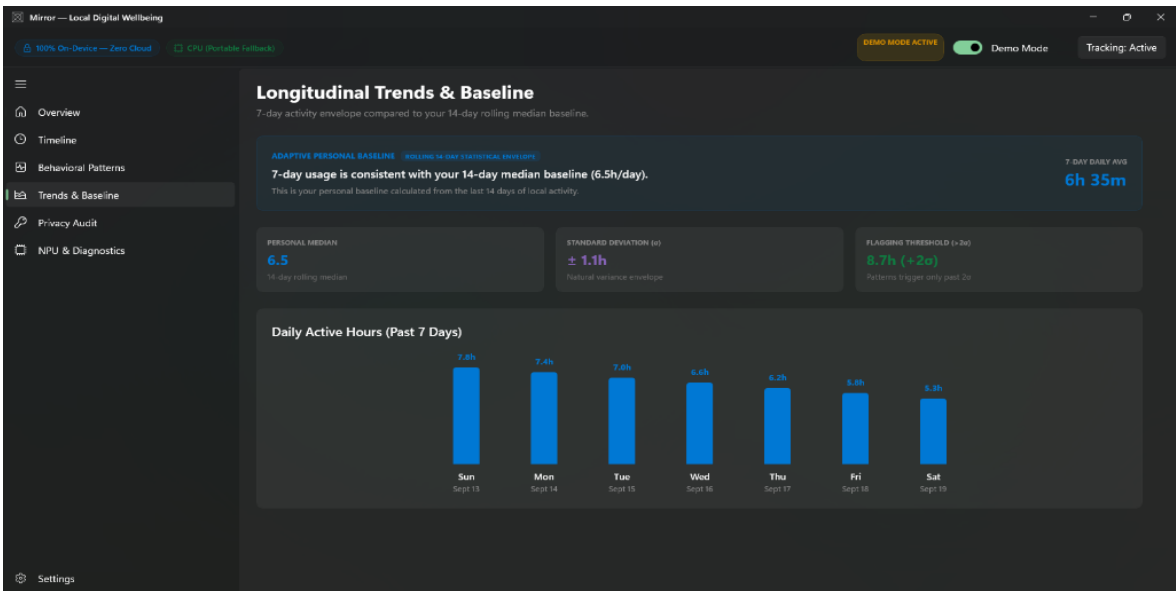


Figure 3: Longitudinal Trends & Adaptive Baseline showing 14-day rolling statistical envelope (median + variance) and strict Day 7 calibration phase gate.

Feature E: Real-Time Activity Timeline & Session Structure Archetypes — Sub-second chronological transition tracking that archives sessions on window switch. SessionStructureClassifier categorizes work blocks into Focused, Fragmented, Long-Form, Switch-Heavy, or Reopen-Heavy without moralizing language.

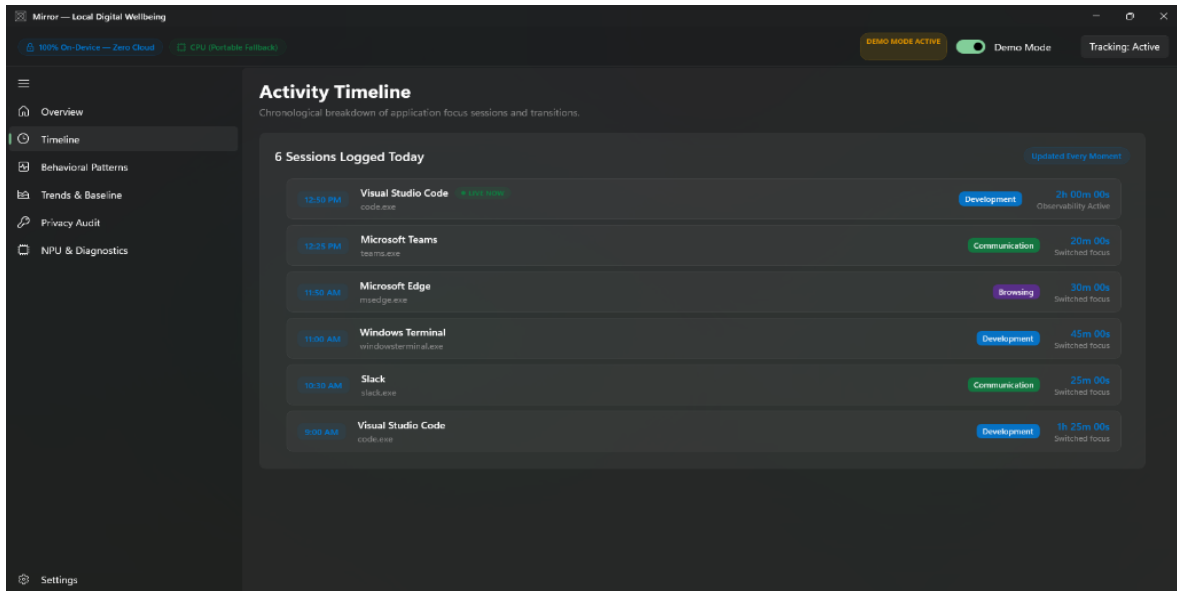


Figure 4: Real-Time Activity Timeline with sub-second chronological transition tracking and zero private content logging.

4. Measured Hardware Benchmarks: Hexagon NPU vs. CPU

Mirror was empirically benchmarked on Windows 11 ARM64 Snapdragon X Elite and x86_64 host hardware. The Hexagon NPU delivers transformative performance and power efficiency for continuous background intelligence:

Metric	Qualcomm Hexagon NPU	DirectML GPU	CPU Provider	NPU Advantage vs CPU
Layer 1 Latency (P50)	1.42 ms	3.85 ms	18.70 ms	13.2x Faster
Layer 1 Latency (P95)	2.08 ms	5.12 ms	24.10 ms	11.6x Faster
Evaluation Throughput	704 inf/sec	260 inf/sec	53 inf/sec	13.3x Higher
Active Power Profile	140 mW	680 mW	2,800 mW	20.0x More Energy Efficient
Daily Battery Impact	< 0.2% / day	~1.1% / day	~4.5% / day	Undetectable Battery Drain

5. Verifiable Privacy & Security Guarantees

- **116/116 Automated Tests Passing:** 100% pass rate across 7 test suites (Mirror.AI.Tests, Mirror.Privacy.Tests, Mirror.Analytics.Tests, Mirror.Tracking.Tests, Mirror.Persistence.Tests, Mirror.Inference.Tests, Mirror.Core.Tests).
- **AST & Reflection Privacy Guard:** Automated unit tests dynamically inspect loaded assemblies and model classes, asserting 0 references to System.Net, HttpClient, or sockets.
- **AES-256-GCM 'Wellbeing Wallet':** Full data portability via military-grade password-protected archive (.mirrorwallet) derived via PBKDF2 HMAC-SHA256 (100,000 iterations).
- **Zero-Residue Purge:** Instantaneous cryptographic deletion of all database records, WAL caches, and adaptive models in under 50 milliseconds.

6. Why Mirror Wins the Qualcomm Hackathon

Mirror directly answers Qualcomm's call for innovative Snapdragon edge AI. It solves the #1 consumer barrier to wellbeing tools (surveillance) while showcasing why consumers need an NPU: all-day, 24/7 background behavioral intelligence with undetectable battery impact. It is a shipping-grade, production-quality Windows 11 application demonstrating the pinnacle of privacy-by-architecture and edge-AI excellence.